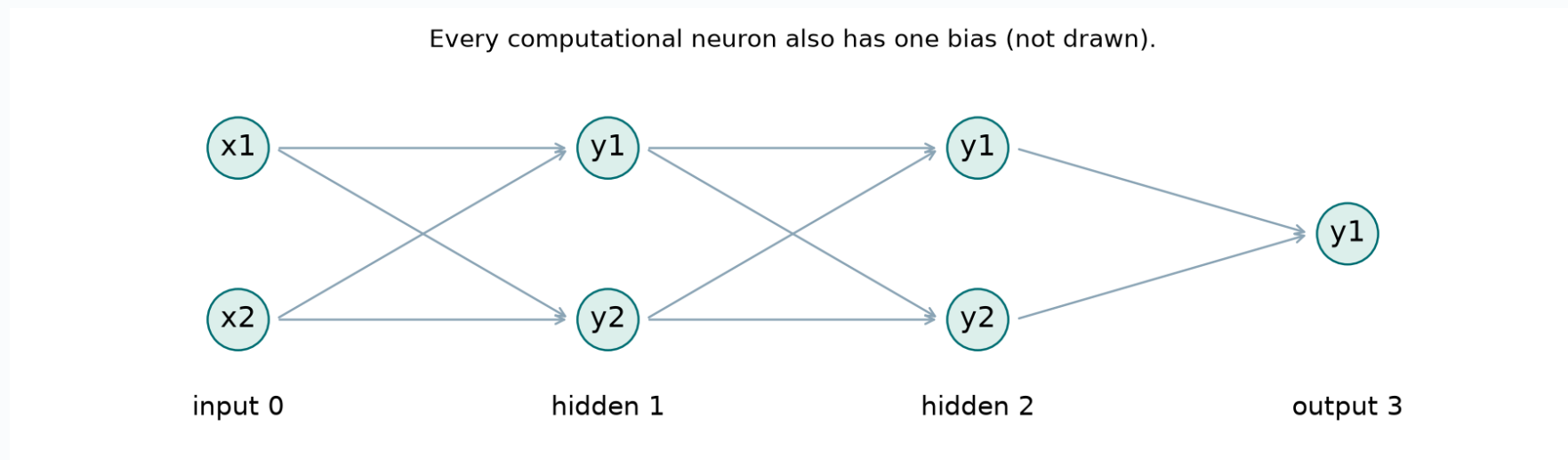


WORKED EXAMPLE

One network for the next five lessons

Architecture and notation follow the backpropagation handout; numbers are study choices.



Input layer 0 has two features. Hidden layers 1 and 2 each have two sigmoid neurons. Layer 3 has one sigmoid output neuron.

Lessons 16-20 reuse the original N3.4 numerical-practice parameters. This lesson checks connections, shapes and counts before the full forward pass.

WORKED EXAMPLE

Read the weight indices as an arrow

The handout writes the source index first and the destination index second.

$w_{i,j}^{(l)}$: source i in layer $l - 1 \rightarrow$ destination j in layer l

$w_{2,1}^{(1)} = 0.3$: $x_2 \rightarrow y_1^{(1)}$

$w_{1,2}^{(1)} = -0.2$: $x_1 \rightarrow y_2^{(1)}$

$b_j^{(l)} = w_{0,j}^{(l)}$

The bias is the weight from constant input 1. We store it separately as b . Do not also add a second constant-input bias if it is already included in b .

z is the weighted sum before activation; y is the activation after sigmoid. Known target labels will be written t .

WORKED EXAMPLE

Count every weight and every bias

Each destination neuron has one weight per source neuron and one bias.

$$\text{layer 1: } 2(2) + 2 = 4 + 2 = 6$$

$$\text{layer 2: } 2(2) + 2 = 4 + 2 = 6$$

$$\text{layer 3: } 2(1) + 1 = 2 + 1 = 3$$

$$\text{weights} = 4 + 4 + 2 = 10, \quad \text{biases} = 2 + 2 + 1 = 5$$

$$\text{total trainable parameters} = 10 + 5 = 15$$

Input values are data, not trainable parameters. The sigmoid function adds no trainable parameter in this network.

WORKED EXAMPLE

Record all original parameter values

Reset to this table whenever a later exercise says original parameters.

layer	w1,1	w1,2	w2,1	w2,2	b1	b2
1	0.1	-0.2	0.3	0.2	0	0.1
2	0.4	-0.3	-0.2	0.2	0.1	-0.1
3	0.3	—	-0.4	—	0.2	—

Layer 3 has only destination 1. Dashes mean those destination-2 connections do not exist; they are not additional zero-valued parameters.

These are chosen teaching values from numerical-practice.md, not numerical values supplied by the handout. The architecture and indexing come from the handout.

WORKED EXAMPLE

Arrange connection weights into a matrix

Rows are sources; columns are destinations. The convention determines the multiplication.

	destination 1	destination 2
source 1	0.1	-0.2
source 2	0.3	0.2

$$W1$$

$W^{(1)}$ has shape 2×2 , $b^{(1)} = [0, 0.1]$

Column 1 contains the two weights feeding hidden neuron 1. Column 2 contains the two weights feeding hidden neuron 2.

WORKED EXAMPLE

Matrix multiplication is a list of dot products

For a single example, place input values in one row.

$$Y^{(0)} = [1, 2], \quad Z^{(1)} = Y^{(0)}W^{(1)} + b^{(1)}$$

$$z_1^{(1)} = 1(0.1) + 2(0.3) + 0 = 0.1 + 0.6 = 0.7$$

$$z_2^{(1)} = 1(-0.2) + 2(0.2) + 0.1 = -0.2 + 0.4 + 0.1 = 0.3$$

$$Z^{(1)} = [0.7, 0.3]$$

Each output entry pairs the input row with one destination column of W . The bias for that destination is added exactly once.

WORKED EXAMPLE

Apply sigmoid to each entry separately

Activation changes the values, not the array shape.

$$Y^{(1)} = \sigma(Z^{(1)})$$

$$y_1^{(1)} = \sigma(0.7) = \frac{1}{1 + e^{-0.7}} \approx 0.668188$$

$$y_2^{(1)} = \sigma(0.3) = \frac{1}{1 + e^{-0.3}} \approx 0.574443$$

$$Y^{(1)} \approx [0.668188, 0.574443], \quad \text{shape } 1 \times 2$$

This is elementwise activation, not a matrix inverse or a sigmoid of the sum of both entries.

Lesson17 continues through the other two layers.

WORKED EXAMPLE

Add a second example as another row

Use the same weights and biases for both examples.

example	x1	x2	z1 in layer1	z2 in layer1
A	1	2	0.7	0.3
B	-1	1	0.2	0.5

$$z_{B,1}^{(1)} = (-1)(0.1) + (1)(0.3) + 0 = 0.2$$

$$z_{B,2}^{(1)} = (-1)(-0.2) + (1)(0.2) + 0.1 = 0.5$$

Input X and output Z1 now both have shape 2 by 2. Each row keeps its own example; columns keep neuron identities.

Bias broadcasting means reuse [0,0.1] on each row. It does not create new learned biases for example B.

WORKED EXAMPLE

Check dimensions before multiplying

Inner dimensions must agree; the outer dimensions give the result shape.

N counts examples; $d(l)$ is the number of neurons in layer l , with $d(0)$ equal to input features.

$$(N \times d_{l-1})(d_{l-1} \times d_l) = N \times d_l$$

$$Z^{(l)} = Y^{(l-1)}W^{(l)} + b^{(l)}$$

stage	Y input	W	bias	Z and Y output
layer1	2 x 2	2 x 2	1 x 2	2 x 2
layer2	2 x 2	2 x 2	1 x 2	2 x 2
layer3	2 x 2	2 x 1	1 x 1	2 x 1

$N = 2$ here counts examples. The final 2 by 1 array has one prediction per example. Increasing N changes activation shapes, not the 15-parameter count.

INDEPENDENT PRACTICE

Your turn: a different small architecture

Use the same row-example and source-to-destination convention.

3 inputs $\rightarrow$ 2 hidden $\rightarrow$ 1 output

1. Count all connection weights, all biases and total trainable parameters.
2. For a batch of four examples, give the shapes of X , $W1$, $b1$, $Z1$, $W2$, $b2$ and $Z2$.
3. For one input $x = (1,0,-1)$, $W1$ has rows $(1,2)$, $(3,4)$, $(5,6)$; $b1 = (0.5,-0.5)$. Compute both entries of $z1$ by scalar multiplication.

Do not apply an activation in question 3: it asks for preactivation z .

WORKED ANSWER

Practice answer: count and check shapes

A larger input width adds connections into every first-layer neuron.

$$\text{weights} = 3(2) + 2(1) = 6 + 2 = 8$$

$$\text{biases} = 2 + 1 = 3, \quad \text{total} = 8 + 3 = 11$$

$$X : 4 \times 3, \quad W^{(1)} : 3 \times 2, \quad b^{(1)} : 1 \times 2$$

$$Z^{(1)} : 4 \times 2, \quad W^{(2)} : 2 \times 1, \quad b^{(2)} : 1 \times 1$$

$$Z^{(2)} : 4 \times 1$$

Four examples share the same 11 parameters. Biases are broadcast over four rows, not trained separately for each row.

WORKED ANSWER

Practice answer: multiply one row by columns

The zero input still has associated weights, even though its contribution is zero here.

$$z_1 = 1(1) + 0(3) + (-1)(5) + 0.5 = 1 + 0 - 5 + 0.5 = -3.5$$

$$z_2 = 1(2) + 0(4) + (-1)(6) - 0.5 = 2 + 0 - 6 - 0.5 = -4.5$$

$$Z^{(1)} = [-3.5, -4.5], \quad \text{shape } 1 \times 2$$

Column 1 was (1,3,5); column 2 was (2,4,6). Multiplying by rows instead would mix up source and destination indices.

Checklist: identify rows and columns, check inner dimensions, add each bias once, and keep parameter count separate from batch size.